Justin Amen N'GBLOGNI

Final-year internship in embodied AI, perception and robotics (M.Eng., 5th year)

Available from March 1 to September 1, 2027

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PROFESSIONAL SUMMARY

Final-year engineering student (ISAE-Supméca), specializing in perception and control of autonomous systems. At Politecnico di Torino, I led a robotic navigation stack from embedded perception to GPU-accelerated predictive control, validated on a real robot. I am seeking a 6-month final-year internship in embodied AI, perception or control, in France or abroad.

EDUCATION

M.Eng. (Diplôme d'Ingénieur) in Mechatronics, Robotics and Automation
ENSIL-ENSCI Limoges, final year at ISAE-Supméca, Paris, France | **September 2024 - September 2027**

Integrated preparatory program, after a scientific baccalaureate with highest honors
ENSIL-ENSCI / École Polytechnique d'Abomey-Calavi, France and Benin | **October 2021 - May 2024**

EXPERIENCE

R&D Intern, Perception and Autonomous Navigation
Politecnico di Torino, DIMEAS Department, Turin, Italy | **May 2026 - July 2026 (3 months)**

- Designed the navigation of a TurtleBot 4 around a two-layer risk map (RGB-D and depth).
- Developed a GPU-accelerated MPPI predictive controller evaluating 5,120 trajectories per cycle at 14 Hz.
- Reduced risk exposure by 44% in forced traversal and by 100% in avoidable zones, at 2 to 9% overhead.
- Modeled the cost arbitration and validated it on the real robot; selected YOLO26 after benchmarking (46 FPS).

Robotics and AI Instructor
Fondation Vallet, EEIA Summer School, Benin | **July 2022 - August 2025 (3 years)**

- Designed and led AI and robotics workshops for about a hundred participants from French-speaking Africa.
- Mentored the capstone project, an autonomous car taken to a working demonstrator, over three editions.

Assembly Line Operator Intern
Stellantis Poissy, Stellantis Group, Poissy, France | **June 2025 - July 2025 (1 month)**

- Assembled Opel Mokka and DS 3 Crossback vehicles in a team, at the takt time of the final line.
- Executed quality checks and safety protocols in a high-volume automotive plant.

PROJECTS

FlyRenov: redesign of a servo-controlled hose winder for a drone
Industrial project, ENSIL-ENSCI × FlyRenov, France | **September 2025 - June 2026**

- Simplified the architecture: 2 motors instead of 4 and a single ESP32 board instead of a CAN network.
- Sized the motor for the worst case (25 Nm, 475 W) and implemented VESC control with a safety ramp.
- Designed a machine one person can carry, using DFMA in 3DEXPERIENCE, with CNC-machined parts.

Autonomous robotic arm playing tic-tac-toe via reinforcement learning
EEIA Summer School, Benin | **July 2025 - August 2025**

- Trained a deep reinforcement learning agent (DQN/PPO) reaching 98% success on complete games.
- Integrated YOLO board detection with real-time embedded inference on a Raspberry Pi 5.

Autonomous car designed end to end
EEIA Summer School, Benin | **July 2021 - August 2024**

- Designed the vehicle: Fusion 360 CAD, 3D printing, embedded electronics and software (ESP32, Raspberry Pi 4).
- Implemented self-driving by imitation learning, moving from distance sensors to camera vision (TensorFlow).

TECHNICAL SKILLS

AI and vision: PyTorch, TensorFlow, OpenCV, YOLO, semantic segmentation, reinforcement learning.
Robotics: ROS 2, Gazebo, Isaac Sim, MPPI, model predictive control, SLAM, Kalman filter, IMU, LiDAR, RGB-D.
Programming: Python, C, C++, CUDA, Linux, Git, ESP32, STM32, RTOS, Raspberry Pi, CAN, I2C, UART.
Design: SolidWorks, Fusion 360, 3DEXPERIENCE, Matlab/Simulink, Simcenter Amesim, KiCad, Altium.

AWARDS, LANGUAGES & INTERESTS

Awards: Fondation Vallet Excellence Scholarship, 2nd National Prize EEIA (2021), MathWorks, Arduino.
Languages: French (native), English (B2), Italian (B1, certified), German (A1).
Interests: Automotive, robotics and AI, chess (1040 Elo), travel, drawing, music, cooking.